

FOND DU LAC, WI, USA

WMO: 726506

Lat: 43.769N

Lon: 88.491W

Elev: 246

StdP: 98.40

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 04840

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-18.7	-26.4	0.3	-20.8	-23.9	0.4	-18.5	11.1	-7.5	10.2	-6.6	3.2	290	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	31.4	23.0	29.9	22.0	28.4	21.2	24.5	29.5	23.5	28.1	22.5	26.9	5.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.0	27.8	22.1	17.3	26.9	21.1	16.2	25.6	75.5	29.1	71.2	28.2	67.2	26.7	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	9.0	8.2	DB	-25.1	33.7	3.1	1.8	-27.4	35.0	-29.2	36.1	-30.9	37.1	-33.2	38.4	
			WB	-25.4	26.2	3.1	1.3	-27.6	27.2	-29.4	27.9	-31.1	28.6	-33.4	29.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-6.9	-5.2	0.8	7.8	14.3	19.6	22.1	20.8	17.0	10.1	3.1	-3.6
	DBStd	11.36	6.70	6.39	6.43	5.07	4.78	3.73	3.17	2.93	4.41	5.05	5.34	6.24
	HDD10.0	2063	526	427	294	101	13	0	0	3	62	214	424	
	HDD18.3	3969	784	659	543	318	144	28	6	11	77	261	457	681
	CDD10.0	1473	0	1	10	33	146	289	375	334	212	65	7	1
	CDD18.3	337	0	0	1	1	18	68	122	87	36	5	0	0
	CDH23.3	2842	0	0	4	22	186	584	1083	634	288	41	0	0
	CDH26.7	784	0	0	0	3	50	163	343	152	67	6	0	0
Wind	WSAvg	3.8	4.2	4.1	4.2	4.5	4.1	3.5	3.2	2.9	3.4	3.9	4.2	4.1
Precipitation	PrecAvg	739	28	23	43	67	81	91	88	91	91	59	49	32
	PrecMax	994	96	57	127	149	342	182	206	206	323	177	146	105
	PrecMin	574	0	2	2	22	12	8	23	10	4	5	1	1
	PrecStd	118	18	16	28	27	56	44	41	47	66	38	32	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	12.8	22.7	26.4	31.1	32.8	34.3	32.1	31.2	27.4	19.9	13.9
		MCWB	6.3	9.5	16.1	16.4	21.4	22.7	24.8	24.0	22.0	20.0	13.6	12.1
	2%	DB	5.4	8.0	17.4	22.5	27.8	30.3	32.0	29.9	28.6	23.8	16.2	9.2
		MCWB	3.0	4.9	12.7	14.1	19.2	22.1	23.7	22.7	21.2	17.7	11.8	7.3
	5%	DB	3.2	5.2	13.3	19.0	25.2	28.5	30.0	28.1	26.8	21.3	13.7	6.3
		MCWB	1.6	2.9	9.1	12.5	17.5	21.1	22.3	21.4	19.8	16.2	10.1	4.2
	10%	DB	2.1	2.8	9.9	16.5	22.7	27.0	28.3	27.0	24.7	18.0	11.2	4.0
		MCWB	0.9	1.1	6.7	10.4	16.4	20.0	21.7	20.6	18.8	14.0	8.2	2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	9.5	16.8	17.9	22.4	25.0	26.5	25.3	23.8	21.3	14.8	12.3
		MCDB	8.4	12.5	22.5	23.5	29.2	30.9	32.2	30.2	28.6	25.9	18.4	13.8
	2%	WB	3.4	5.3	13.6	15.5	20.3	23.6	24.9	23.9	22.2	19.1	12.6	7.7
		MCDB	5.0	7.7	16.7	21.0	25.6	28.7	30.1	28.3	26.7	22.5	15.5	8.8
	5%	WB	1.9	3.2	9.8	13.3	18.9	22.3	23.8	22.7	21.1	16.7	10.5	4.8
		MCDB	3.2	5.2	13.0	18.4	23.9	27.0	28.3	26.9	25.1	20.2	13.3	6.1
	10%	WB	0.7	1.4	6.6	11.0	17.3	20.8	22.8	21.5	19.9	14.6	8.1	2.6
		MCDB	2.0	2.8	9.8	15.9	21.6	25.2	27.0	25.4	23.7	18.1	10.6	4.1
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.3	9.0	10.4	10.9	10.6	10.8	10.3	11.1	9.7	8.2	7.3
		MCDBR	7.8	9.1	12.5	14.8	13.8	12.4	12.1	11.5	12.6	13.0	11.8	8.6
		MCWBR	6.7	6.9	8.5	8.9	7.5	6.5	5.9	5.9	6.5	7.8	8.4	7.2
	5% WB	MCDBR	7.5	8.6	11.7	13.6	12.1	11.0	10.4	10.0	10.6	11.2	10.9	8.0
		MCWBR	6.7	6.8	8.6	9.0	7.3	6.7	6.0	5.8	6.3	8.0	8.7	7.3
Clear-Sky Solar Irradiance	taub	0.275	0.320	0.338	0.380	0.404	0.417	0.408	0.402	0.372	0.334	0.298	0.276	
	taud	2.388	2.240	2.348	2.279	2.263	2.265	2.294	2.301	2.378	2.477	2.507	2.436	
	Ebn at Noon	861	862	899	883	869	855	859	852	852	846	830	825	
	Edh at Noon	76	106	109	127	132	133	128	122	105	83	68	66	
All-Sky Solar Radiation	RadAvg	1.61	2.68	3.80	4.55	5.42	6.00	6.12	5.24	4.23	2.69	1.73	1.24	
	RadStd	0.21	0.27	0.43	0.43	0.43	0.43	0.28	0.33	0.34	0.29	0.15	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-1.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page